Machine Learning

Gradient Descent, Part 2

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Exercise

- **Given the following objective function, what is the gradient of f at an arbitrary point β^0? And, what is the update rule for an arbitrary model parameter β_j?**

$$f(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

Gradient

- $\nabla f =$

Update Rule

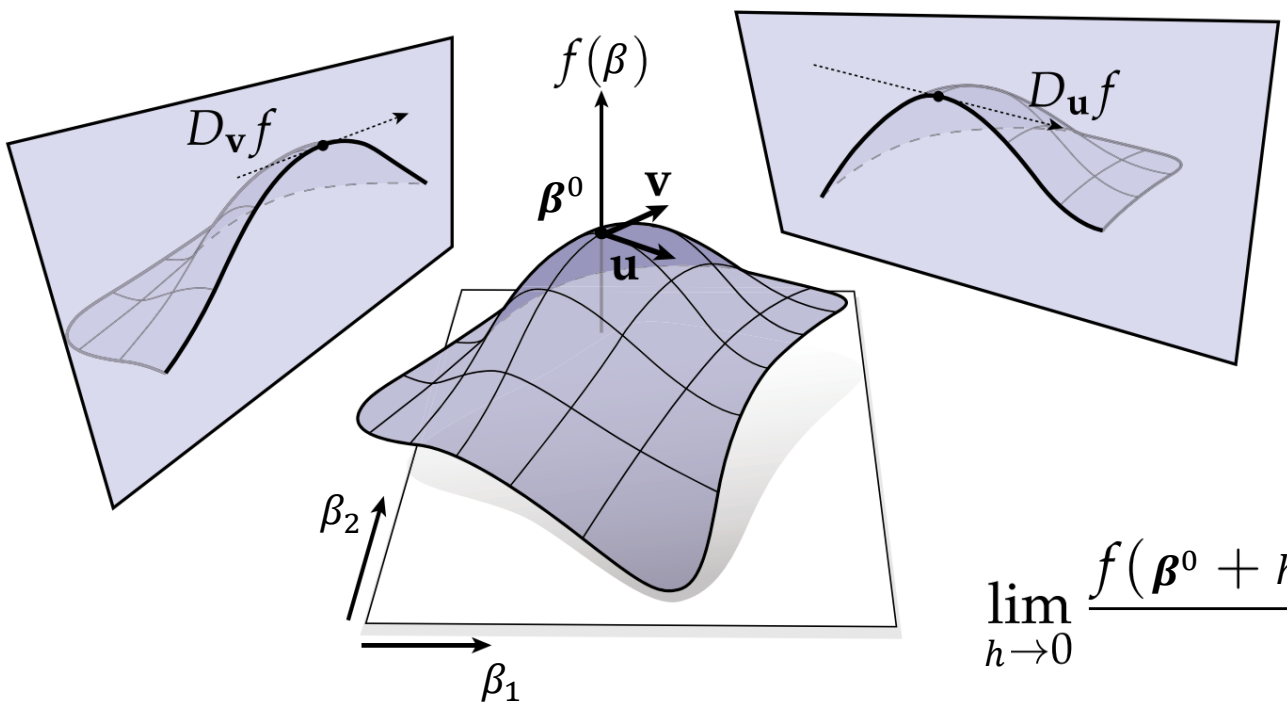
- $\beta_j^{t+1} =$

Property of Gradient

- It points to a direction in which a function increases most quickly
- To verify this, let's take a directional derivative of a function at a specific point

Directional Derivative

- **Estimate the difference of a function according to a direction**
  - Take a derivative at a point $\boldsymbol{x_0}$ along some direction (unit vector $\mathbf{u}$).
  - Then just apply the usual derivative, in terms of the variable h.

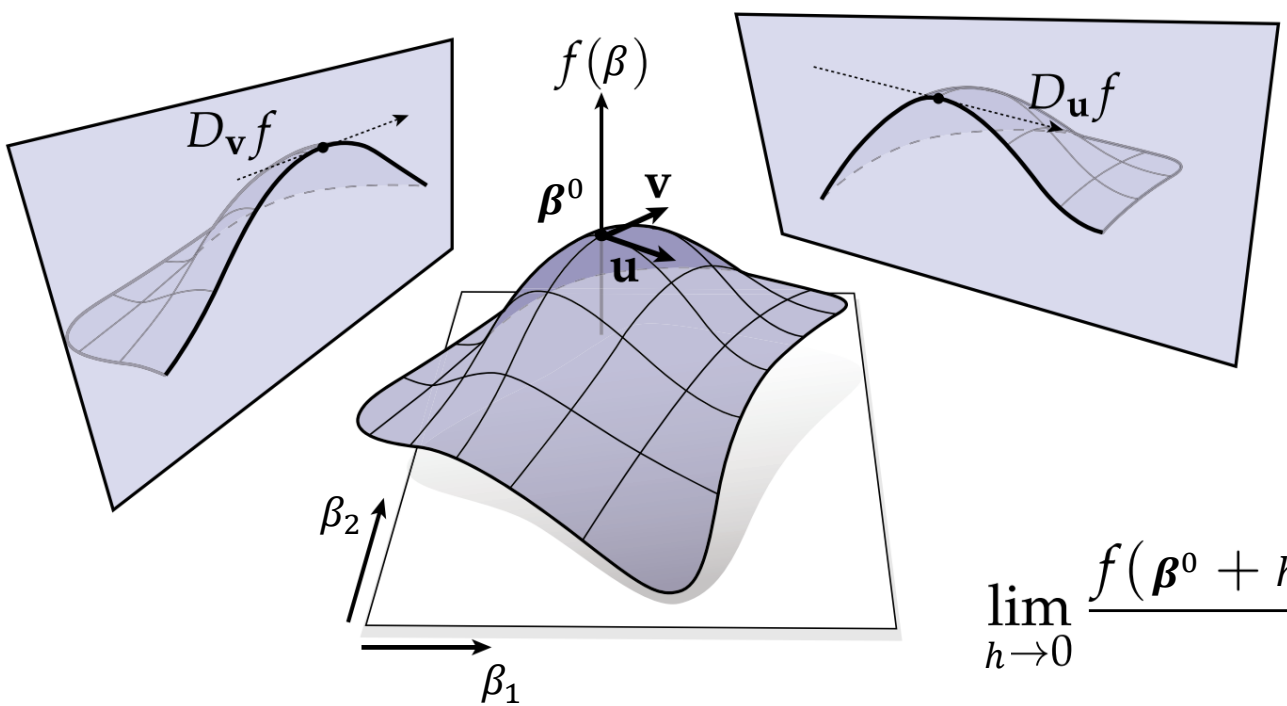


take a small step along u

$$\lim_{h \to 0} \frac{f(\boldsymbol{\beta}^0 + h\mathbf{u}) - f(\boldsymbol{\beta}^0)}{h}$$

Directional Derivative

- **Step by step**
  - Suppose we fix a specific point $\boldsymbol{\beta}^0$ and a direction $\mathbf{u} = [u_1, u_2]$
  - Then, the curve $D_{\mathbf{u}}f$ is a function of h : $f\big(\beta_1(h), \beta_2(h)\big)$ where $\beta_1(h) = \beta_1^0 + h \cdot u_1$ and $\beta_2(h) = \beta_2^0 + h \cdot u_2$



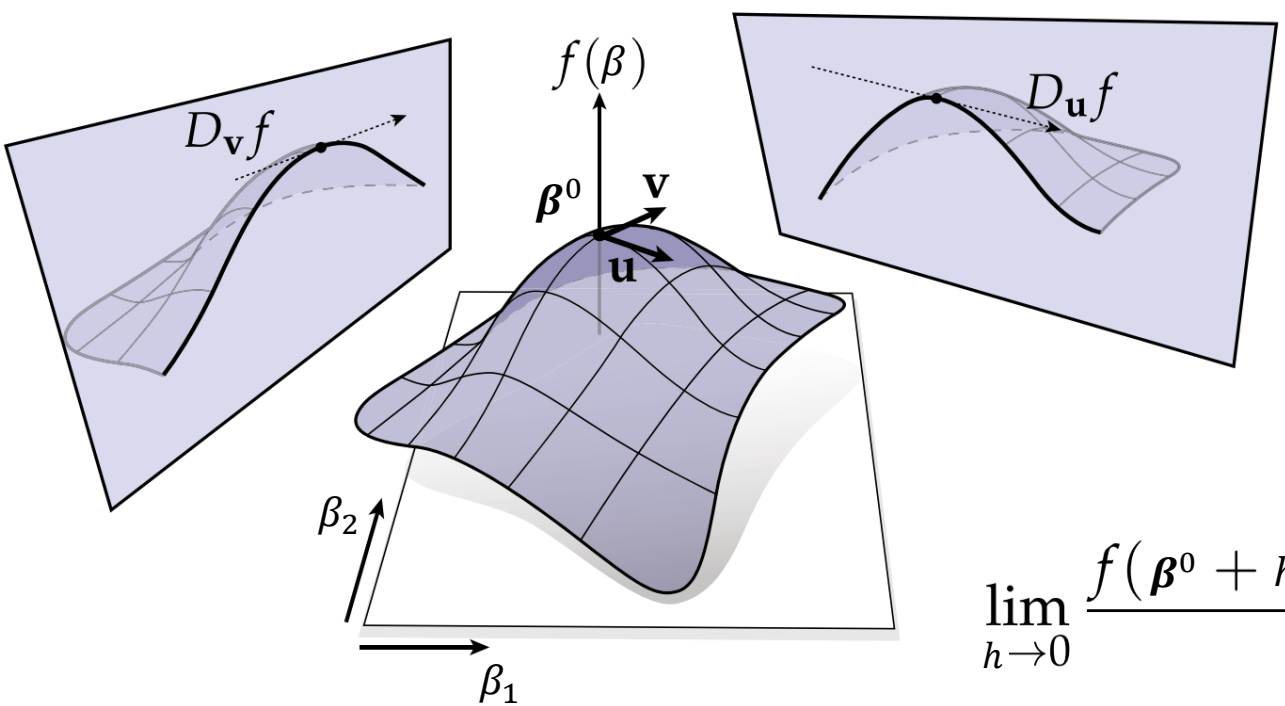
take a small step along u

$$\lim_{h \to 0} \frac{f(\boldsymbol{\beta}^0 + h\mathbf{u}) - f(\boldsymbol{\beta}^0)}{h}$$

Directional Derivative

$$f(\beta) = f\big(\beta_1(h), \beta_2(h)\big)$$
$$\beta_1(h) = \beta_1^0 + h \cdot u_1$$
$$\beta_2(h) = \beta_2^0 + h \cdot u_2$$

- **Step by step**
  - Now let's compute $\dfrac{df}{\mathrm{d}h}$
  - By applying the chain rule, $\dfrac{\mathrm{d}f}{\mathrm{d}h} = \dfrac{\partial f}{\partial \beta_1}\dfrac{\mathrm{d}\beta_1}{\mathrm{d}h} + \dfrac{\partial f}{\partial \beta_2}\dfrac{\mathrm{d}\beta_2}{\mathrm{d}h}$

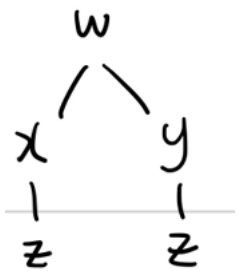


take a small step along u

$$\lim_{h \to 0} \frac{f(\boldsymbol{\beta}^0 + h\mathbf{u}) - f(\boldsymbol{\beta}^0)}{h}$$

Appendix

- **Derivative rule of the situation:**



$$\frac{dw}{dz} = \frac{\partial w}{\partial x}\frac{dx}{dz} + \frac{\partial w}{\partial y}\frac{dy}{dz}$$

why?

$$\frac{dw}{dz} \triangleq \lim_{\Delta z \to 0} \frac{w(x(z+\Delta z), y(z+\Delta z)) - w(x(z), y(z))}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{w(x(z+\Delta z), y(z+\Delta z)) - w(x(z), y(z+\Delta z))}{\Delta z} + \frac{w(x(z), y(z+\Delta z)) - w(x(z), y(z))}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{w(x(z+\Delta z), y(z+\Delta z)) - w(x(z), y(z+\Delta z))}{x(z+\Delta z) - x(z)} \frac{x(z+\Delta z) - x(z)}{\Delta z} + \frac{w(x(z), y(z+\Delta z)) - w(x(z), y(z))}{y(z+\Delta z) - y(z)} \frac{y(z+\Delta z) - y(z)}{\Delta z}$$

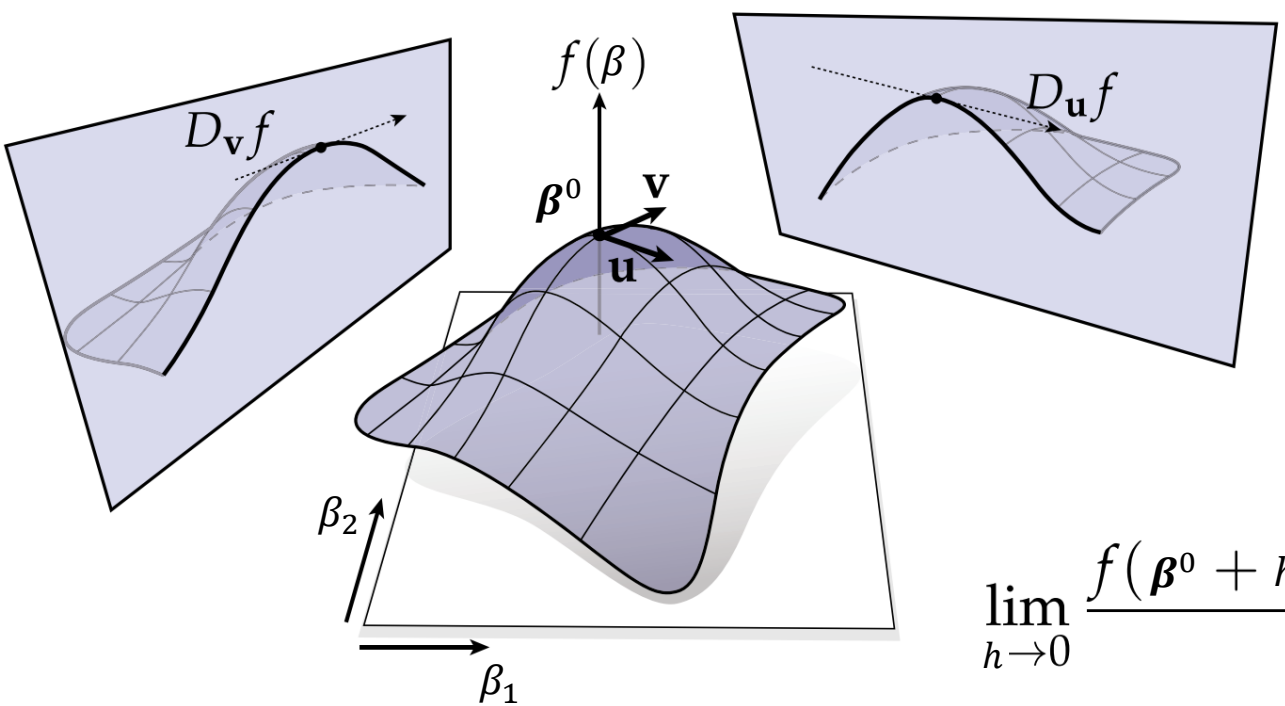
$$= \frac{\partial w}{\partial x}\frac{dx}{dz} + \frac{\partial w}{\partial y}\frac{dy}{dz}$$

Directional Derivative

$$f(\beta) = f\big(\beta_1(h), \beta_2(h)\big)$$
$$\beta_1(h) = \beta_1^0 + h \cdot u_1$$
$$\beta_2(h) = \beta_2^0 + h \cdot u_2$$

- **Step by step**
  - $$\frac{\mathrm{d}f}{\mathrm{d}h} = \frac{\partial f}{\partial \beta_1}\frac{\mathrm{d}\beta_1}{\mathrm{d}h} + \frac{\partial f}{\partial \beta_2}\frac{\mathrm{d}\beta_2}{\mathrm{d}h} = \frac{\partial f}{\partial \beta_1} \cdot u_1 + \frac{\partial f}{\partial \beta_2} \cdot u_2$$

$$= \left[\frac{\partial f}{\partial \beta_1}, \frac{\partial f}{\partial \beta_2}\right] \cdot [u_1, u_2] = \boldsymbol{\nabla f} \cdot \mathbf{u}$$



take a small step along u

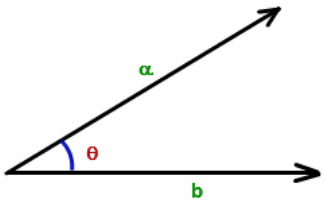
$$\lim_{h \to 0} \frac{f(\boldsymbol{\beta}^0 + h\mathbf{u}) - f(\boldsymbol{\beta}^0)}{h}$$

Directional Derivative

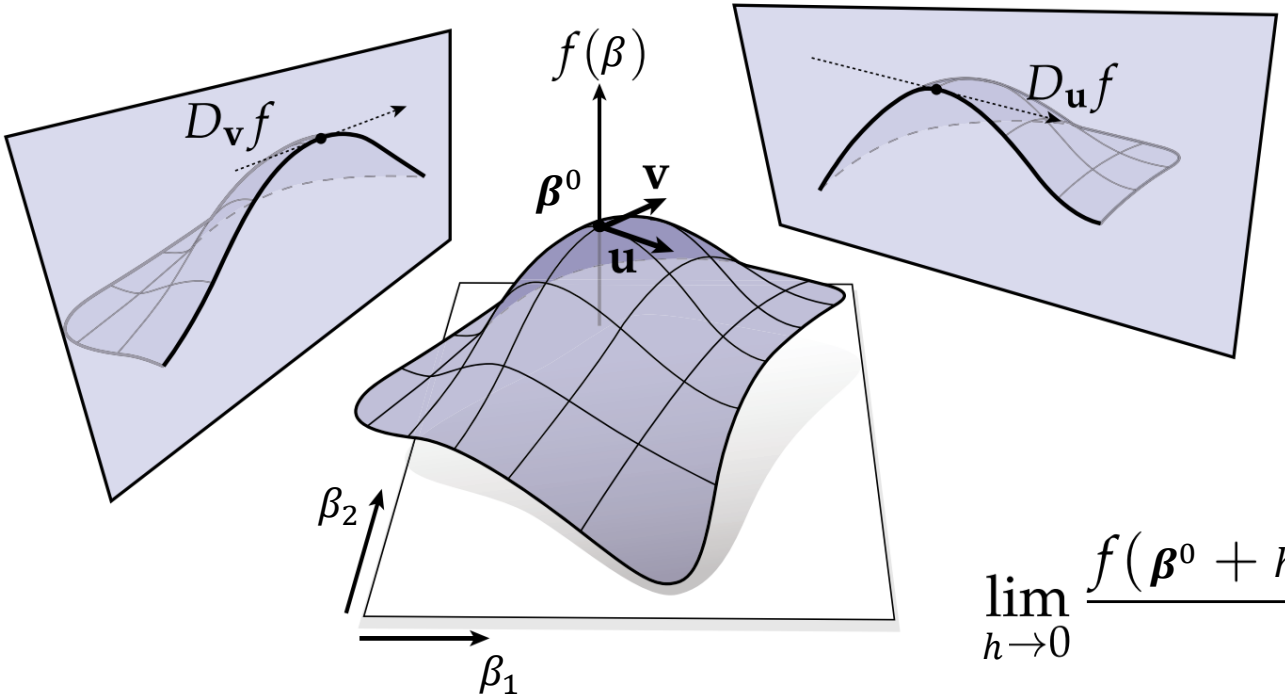
- **The difference of f according to a direction u is: $\nabla f \cdot \mathbf{u}$**

- **Question: when $\nabla f \cdot \mathbf{u}$ is maximized?**

  => Both ∇f and $\mathbf{u}$ point to the same direction!



a.b = |a| |b| Cosθ



take a small step along u

$$\lim_{h \to 0} \frac{f(\boldsymbol{\beta}^0 + h\mathbf{u}) - f(\boldsymbol{\beta}^0)}{h}$$

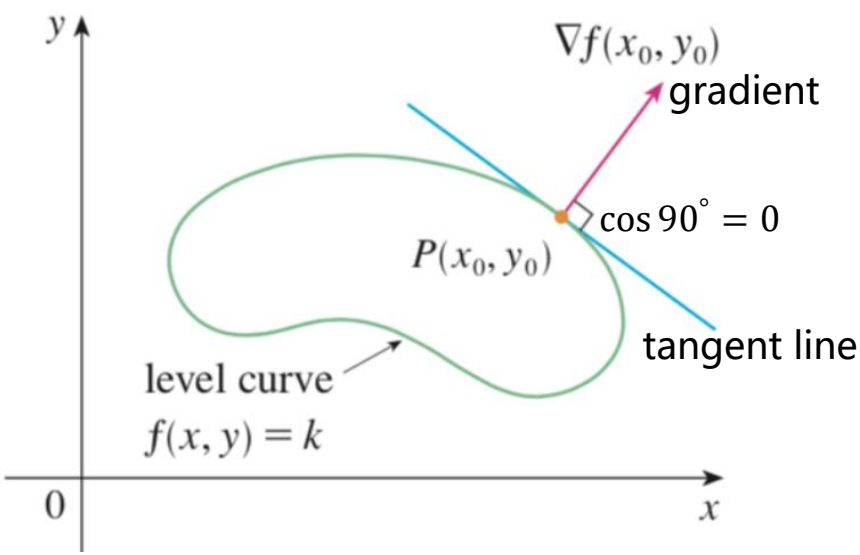
Conclusions

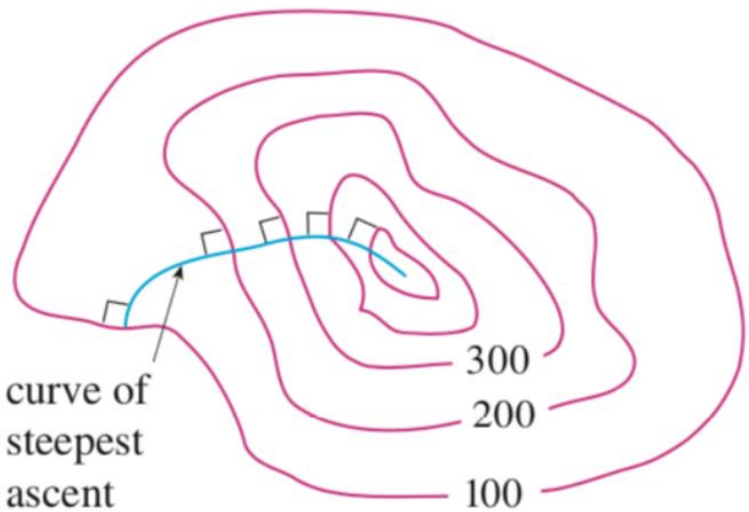
- The gradient vector points to a direction in which a function increases most quickly
- To verify this, we take a directional derivative of a function (방향에 따른 함수의 변화량) at a specific point and a specific direction
- As a result, we have a result of $\nabla f \cdot \mathbf{u}$ regardless of how we choose the specific point and direction
- To maximize $\nabla f \cdot \mathbf{u}$, the direction $\mathbf{u}$ must point to the same direction with the gradient vector ∇f !

Appendix

❑ Another approach

- ❑ Gradients are **orthogonal to tangent lines of level curves** (and level surfaces), which means that it points to a most quickly increasing direction
  - Let's think about a directional derivative of f on an arbitrary point of a line curve along the direction of the line curve.
  - Then the result $\frac{df}{dh}$ will always 0 since there is no change of f on the line curve.
  - $\nabla f \cdot \mathrm{u}$ is always 0, but u is not a zero vector. So, to make the result zero, the two vectors must be orthogonal, which makes $\cos\theta = 0$.





Thank You